

CHAPTER II

Pair of Straight Lines

Consider the general equation of second degree in x and y :

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \quad (1)$$

To remove the first degree terms from (1), we must shift the origin to (α, β) , where

$$\alpha = \frac{fh - bg}{ab - h^2} \text{ and } \beta = \frac{gh - af}{ab - h^2}$$

and to remove the product term xy from (1), we must rotate the reference axes through an angle

$$\theta = \frac{1}{2} \tan^{-1} \left(\frac{2h}{a - b} \right).$$

Homogeneous Quadratic Equations

Consider the equation $ax^2 + 2hxy + by^2 = 0$ (2)

If $b \neq 0$, divide (2) by bx^2 to get

$$\left(\frac{y}{x} \right)^2 + 2 \frac{h}{b} \frac{y}{x} + \frac{a}{b} = 0 \quad (3)$$

Let m_1, m_2 be the roots of (3). Then we have

$$m_1 + m_2 = -\frac{2h}{b} \text{ and } m_1 m_2 = \frac{a}{b} \quad (4)$$

Since (3) is equivalent to $\left(\frac{y}{x} - m_1 \right) \left(\frac{y}{x} - m_2 \right) = 0$, the lines represented by (3) are given by

$\frac{y}{x} - m_1 = 0$ and $\frac{y}{x} - m_2 = 0$, i.e., $y - m_1 x = 0$ and $y - m_2 x = 0$ both of which pass through the origin.

Thus, the homogeneous quadratic equation (2) represents a pair of straight lines, real or imaginary, passing through the origin.

Note: An n -th degree homogeneous equation represents n straight lines, real or imaginary, passing through the origin.

An equation of the form

$$a_0 y^n + a_1 y^{n-1} x + a_2 y^{n-2} x^2 + \dots + a_n x^n = 0 \quad (5)$$

in which the sum of the powers of x and y in every term is the same (say, n) is called a *homogeneous equation* (of degree n).

Angle between the lines represented by the equation (2)

We know the lines represented by (2) are $y - m_1 x = 0$ and $y - m_2 x = 0$. Also, in this case, we

know $m_1 + m_2 = -\frac{2h}{b}$ and $m_1 m_2 = \frac{a}{b}$.

Let θ be the angle between the straight lines $y - m_1 x = 0$ and $y - m_2 x = 0$. Then

$$\tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2} = \frac{\sqrt{(m_1 + m_2)^2 - 4m_1 m_2}}{1 + m_1 m_2} = \frac{\sqrt{\left(-\frac{2h}{b}\right)^2 - 4\frac{a}{b}}}{1 + \frac{a}{b}} = \frac{2\sqrt{h^2 - ab}}{a + b}$$

The following cases may arise:

(I) If the lines $y - m_1 x = 0$ and $y - m_2 x = 0$ are perpendicular, then $a + b = 0$.

(II) If the lines $y - m_1 x = 0$ and $y - m_2 x = 0$ are parallel, then $h^2 = ab$.

General equation of second degree

Find the condition that the general equation of second degree (1) may represent a pair of straight lines not passing through the origin. (Ref. S. L. Loney, page 95)

The equation (1) will represent a pair of straight lines if

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

or, if $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$.

Angle between the lines represented by (1)

If (1) represents two straight lines $lx + my + n = 0$ and $l'x + m'y + n' = 0$ and if θ be the angle between the straight lines, then we can show that

$$\tan \theta = \frac{2\sqrt{h^2 - ab}}{a + b}.$$

The following cases may arise:

(I) If the lines $lx + my + n = 0$ and $l'x + m'y + n' = 0$ are perpendicular, then $\Delta = 0$ and $a + b = 0$.

(II) If the lines $lx + my + n = 0$ and $l'x + m'y + n' = 0$ are parallel, then $\Delta = 0$ and $h^2 = ab$.

(III) If the lines $lx + my + n = 0$ and $l'x + m'y + n' = 0$ coincide, then

$$\Delta = 0 \text{ and } gh = af, hf = bg \text{ and } fg = ch.$$

Bisectors of the angles between the lines

Let (α, β) be the point of intersection of the lines represented by (1). If we transfer the origin to (α, β) without rotation, the first degree terms will vanish. In this case, the transformed equation will be a homogeneous equation of the form (2).

If (α, β) be the origin, then the equation of the straight lines bisecting the angle between the

straight lines given by (1) are $\frac{x^2 - y^2}{a - b} = \frac{xy}{h}$. (Ref. S. L. Loney, page 92)

If we go back to our old axes, then the equation of the bisectors of the angle between the lines represented by (1) are

$$\frac{(x - \alpha)^2 - (y - \beta)^2}{a - b} = \frac{(x - \alpha)(y - \beta)}{h}.$$

Lines joining the origin to the intersection of a curve and a line

Consider the second degree equation

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \quad (1)$$

and the equation of a straight line $lx + my = n$ (6)

$$\text{Now, } lx + my = n \Rightarrow \frac{l}{n}x + \frac{m}{n}y = 1 \quad (7)$$

Let the straight lines meet the curve at P and Q . It requires find the equation of the pair of lines OP and OQ .

From (1) and (7), we get

$$ax^2 + 2hxy + by^2 + 2(gx + fy)\left(\frac{lx + my}{n}\right) + c\left(\frac{lx + my}{n}\right)^2 = 0$$

$$\text{which gives } Ax^2 + 2Hxy + By^2 = 0 \quad (8)$$

$$\text{where } A = \frac{an^2 + 2gnl + cl^2}{n^2}, H = \frac{2(hn^2 + gmn + fnl + lm)}{n^2}, \text{ and } B = \frac{bn^2 + 2fmn + cm^2}{n^2},$$

which is a homogeneous equation of the second degree. It represents a pair of straight lines passing through the origin. Moreover, the points P and Q satisfy (8) because they satisfy (1) and (7).

Worked out problems

1. Prove that the equation $x^2 + 6xy + 9y^2 + 4x + 12y - 5 = 0$ represents a pair of parallel lines.

Solution: We have $x^2 + 6xy + 9y^2 + 4x + 12y - 5 = 0$

$$\Rightarrow x^2 + (6y + 4)x + (9y^2 + 12y - 5) = 0$$

which is quadratic in x .

$$\text{So, } x = \frac{-(6y + 4) \pm \sqrt{(6y + 4)^2 - 4 \times 1 \times (9y^2 + 12y - 5)}}{2} = \frac{-6y - 4 \pm 6}{2} = -3y - 2 \pm 3$$

$$\Rightarrow x + 3y + 5 = 0 \text{ and } x + 3y - 1 = 0$$

Since the lines differ only with constant terms, the lines represented by the given equation are parallel.

2. Show that the equation $2x^2 + 7xy + 3y^2 + 8x + 14y + 8 = 0$ represents a pair of straight lines and find them. Find their point of intersection and the angle made by the lines.

Solution: We have $2x^2 + 7xy + 3y^2 + 8x + 14y + 8 = 0$ (1)

Compare (1) to $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, we get

$$a = 2, b = 3, c = 8, h = \frac{7}{2}, g = 4, f = 7$$

Equation (1) will represent a pair of straight lines if

$$abc + 2fgh - af^2 - bg^2 - ch^2 = 0 \quad (2)$$

Putting the above values in (2), we get

$$2 \times 3 \times 8 + 2 \times 7 \times 4 \times \frac{7}{2} - 2 \times 49 - 3 \times 16 - 8 \times \frac{49}{4} = 0$$

$$\Rightarrow 48 + 196 - 98 - 48 - 98 = 0 \Rightarrow 0 = 0$$

Therefore, the given equation represents a pair of straight lines.

Rewrite the given equation in the form as

$$2x^2 + 7xy + 3y^2 + 8x + 14y + 8 = 0 \Rightarrow 2x^2 + (7y + 8)x + (3y^2 + 14y + 8) = 0$$

which is quadratic in x .

Therefore, by general quadratic formula, we get

$$x = \frac{-(7y + 8) \pm \sqrt{(7y + 8)^2 - 4(2)(3y^2 + 14y + 8)}}{(2)(2)} = \frac{1}{4}(-7y - 8) \pm \sqrt{25y^2}$$

$$\Rightarrow 4x + 7y + 8 = \pm 5y \Rightarrow 4x + 2y + 8 = 0 \text{ or } 4x + 12y + 8 = 0$$

$$\Rightarrow 2x + y + 4 = 0 \text{ or } x + 3y + 2 = 0$$

which are the desired equations.

Solving them, we get $x = -2$ and $y = 0$. So, the point of intersection is $(-2, 0)$.

Finally, let θ be the angle between the straight lines represented by (1). Then

$$\tan \theta = \frac{2\sqrt{h^2 - ab}}{a + b} = \frac{2\sqrt{(49/4) - 2 \times 3}}{2 + 3} = 1$$

Therefore, $\theta = \tan^{-1}(1) = \pi/4$.

3. Find the values of k so that the equation $3x^2 + 10xy + 8y^2 - kx - 26y + 21 = 0$ represents a pair of straight lines. Also, find the angle between the lines.

Solution: We have $3x^2 + 10xy + 8y^2 - kx - 26y + 21 = 0$ (1)

Consider the general equation of second degree

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \quad (2)$$

Comparing (1) with (2), we get $a = 3, b = 8, c = 21, h = 5, g = -\frac{k}{2}, f = -13$.

The given equation (1) will represent a pair of straight lines if

$$abc + 2fgh - af^2 - bg^2 - ch^2 = 0 \quad (3)$$

Putting the above values in (3), we get

$$\begin{aligned} 3 \times 8 \times 21 + 2(-13) \times 5 - 3(-13)^2 - 8\left(-\frac{k}{2}\right)^2 - 21 \times 25 &= 0 \\ \Rightarrow 504 + 65k - 507 - 2k^2 - 525 &= 0 \\ \Rightarrow 2k^2 - 65k + 528 &= 0 \\ \Rightarrow 2k^2 - 33k - 32k + 528 &= 0 \\ \Rightarrow k(2k - 33) - 16(2k - 33) &= 0 \\ \Rightarrow (k - 16)(2k - 33) &= 0 \\ \Rightarrow k = 16 \text{ or } k = \frac{33}{2} \end{aligned}$$

Let θ be the angle between the straight lines represented by (1). Then

$$\tan \theta = \frac{2\sqrt{h^2 - ab}}{a + b} = \frac{2\sqrt{25 - 3 \times 8}}{3 + 8} = \frac{2}{11}$$

Therefore, $\theta = \tan^{-1}\left(\frac{2}{11}\right)$.

4. Prove that the equation $6x^2 + 7xy - 5y^2 - 22x - 41y - 8 = 0$ represents a pair of straight lines. Also, find their point of intersection and the angle between them.

Solution: (First Part) Left as an Exercise.

(Second Part) We have $6x^2 + 7xy - 5y^2 - 22x - 41y - 8 = 0$ (1)

Consider the general equation of second degree

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \quad (2)$$

Comparing (1) with (2), we get $a = 6, b = -5, c = -8, h = \frac{7}{2}, g = -11, f = -\frac{41}{2}$.

Let (α, β) be the point of intersection made by the lines. Then we know

$$\begin{aligned} \alpha &= \frac{fh - bg}{ab - h^2} = \frac{\frac{7}{2}\left(-\frac{41}{2}\right) - (-5)(-11)}{6(-5) - \left(\frac{7}{2}\right)^2} = \frac{-\frac{287}{4} - 55}{-30 - \frac{49}{4}} = \frac{507}{169} = 3, \\ \text{and } \beta &= \frac{gh - af}{ab - h^2} = \frac{(-11)\frac{7}{2} - 6\left(-\frac{41}{2}\right)}{6(-5) - \left(\frac{7}{2}\right)^2} = \frac{\frac{77}{2} + 123}{-\frac{169}{4}} = -2. \end{aligned}$$

So, the required point of intersection is $(3, -2)$.

(Third Part) Let θ be the angle between the straight lines represented by (1). Then

$$\tan \theta = \frac{2\sqrt{\left(\frac{7}{2}\right)^2 - 6(-5)}}{6-5} = 2\sqrt{\frac{49}{4} + 30} = 2\sqrt{\frac{169}{4}} = \frac{2 \times 13}{2} = 13$$

Therefore, $\theta = \tan^{-1}(13)$.

5. If the equation $ax^2 + 3xy - 2y^2 - 5x + 5y + c = 0$ represents a pair of straight lines perpendicular to each other, find the values of a and c .

Solution. Given $ax^2 + 3xy - 2y^2 - 5x + 5y + c = 0$ (1)

Since equation (1) represents a pair of straight lines, we can write

$$\begin{aligned} a(-2)c + 2 \times \frac{5}{2} \times \left(-\frac{5}{2}\right) \times \frac{3}{2} - a\left(\frac{5}{2}\right)^2 - (-2)\left(-\frac{5}{2}\right)^2 - c\left(\frac{3}{2}\right)^2 &= 0 \\ \Rightarrow -2ac - \frac{75}{4} - \frac{25}{4}a + \frac{50}{4} - \frac{9}{4}c &= 0 \\ \Rightarrow -2ac - \frac{25}{4}a - \frac{9}{4}c - \frac{25}{4} &= 0 \end{aligned} \quad (2)$$

Also, since the lines represented by (1) are perpendicular, we have $a - 2 = 0 \Rightarrow a = 2$.

Hence, from (2), we get

$$-2 \times 2c - \frac{25}{4} \times 2 - \frac{9}{4}c - \frac{25}{4} = 0 \Rightarrow -\frac{16+9}{4}c - \frac{75}{4} = 0 \Rightarrow c = -3.$$

6. Prove that the straight lines joining the origin to the points of intersection of the straight line $x - y = 2$ and the curve $5x^2 + 12xy - 8y^2 + 8x - 4y + 12 = 0$ make equal angles with the axes.

Solution: The equation to the straight lines joining the origin to the points of intersection of the straight line $x - y = 2$ and the curve $5x^2 + 12xy - 8y^2 + 8x - 4y + 12 = 0$ is

$$\begin{aligned} 5x^2 + 12xy - 8y^2 + (8x - 4y)\frac{x-y}{2} + 12\left(\frac{x-y}{2}\right)^2 &= 0 \\ \Rightarrow y^2 &= 4x^2 \end{aligned}$$

which is a homogeneous equation and therefore it represents two straight lines passing through the origin. Therefore, the equations of the lines are $y = 2x$ and $y = -2x$ which are equally inclined to the coordinate axes.

7. Find the value of k so that the straight lines joining the origin to the points of intersection of the straight line $y - x - k = 0$ and the curve $x^2 + y^2 + 4x - 6y - 36 = 0$ are at right angles.

Solution: The equation to the straight lines joining the origin to the points of intersection of the straight line $y - x - k = 0$ and the curve $x^2 + y^2 + 4x - 6y - 36 = 0$ is

$$\begin{aligned} x^2 + y^2 + 2(2x - 3y)\frac{y-x}{k} - 36\left(\frac{y-x}{k}\right)^2 &= 0 \\ \Rightarrow k^2(x^2 + y^2) + 2k(2x - 3y)(y - x) - 36(y^2 - 2xy + x^2) &= 0 \\ \Rightarrow k^2(x^2 + y^2) + 2k(5xy - 2x^2 - 3y^2) - 36(x^2 - 2xy + y^2) &= 0 \\ \Rightarrow (k^2 - 4k - 36)x^2 + (10k + 72)xy + (k^2 - 6k - 36)y^2 &= 0 \end{aligned}$$

which is a homogeneous equation and therefore it represents two straight lines passing through the origin. If the lines are at right angles, then

$$(k^2 - 4k - 36) + (k^2 - 6k - 36) = 0 \Rightarrow 2k^2 - 10k - 72 = 0 \Rightarrow k = -4, 9$$

Finding the sum and product of the roots of a cubic equation

Consider the general form of the cubic equation

$$ax^3 + bx^2 + cx + d = 0,$$

where a, b, c, d are constants and $a \neq 0$. Then the following statements are true:

- (i) the sum of the roots of this equation is $-\frac{b}{a}$,
- (ii) the product of the roots of this equation is $-\frac{d}{a}$,
- (iii) if p, q, r are the roots of this equation, then sum of the product of the roots, taken two at a time, is given by $pq + qr + pr = \frac{c}{a}$.

Example

Consider the cubic equation $9x^3 - 6x^2 - 3x - 2 = 0$. Then

- (i) the sum of the roots of this equation is $\frac{2}{3}$;
- (ii) the product of the roots of this equation is $\frac{2}{9}$;
- (iii) sum of the product of the roots, taken two at a time, is $-\frac{1}{3}$.

8. If two of the three straight lines given by the equation $ax^3 + 3bx^2y + 3cxy^2 + dy^3 = 0$ are at right angles, prove that $a^2 + 3ac + 3bd + d^2 = 0$.

Solution: Given $ax^3 + 3bx^2y + 3cxy^2 + dy^3 = 0$ (1)

This is a homogeneous equation in x and y of degree 3. Therefore, (1) represents three straight lines through the origin.

The slopes of these lines are given by the equations

$$\begin{aligned} d\left(\frac{y}{x}\right)^3 + 3c\left(\frac{y}{x}\right)^2 + 3b\left(\frac{y}{x}\right) + a &= 0 \\ \Rightarrow dm^3 + 3cm^2 + 3bm + a &= 0 \text{ for } \frac{y}{x} = m \end{aligned} \quad (2)$$

Let the roots of (2) be p, q, r . Then $pqr = -\frac{a}{d}$ (3)

Since two of the three straight lines are at right angles, $pq = -1$ (4)

From (3) and (4), we get $r = \frac{a}{d}$

Since r is a root of (2), we have

$$\begin{aligned} d\left(\frac{a}{d}\right)^3 + 3c\left(\frac{a}{d}\right)^2 + 3b\left(\frac{a}{d}\right) + a &= 0 \\ \Rightarrow \frac{a^3}{d^2} + 3c\frac{a^2}{d^2} + 3b\frac{a}{d} + a &= 0 \\ \Rightarrow a^3 + 3a^2c + 3abd + ad^2 &= 0 \\ \Rightarrow a^2 + 3ac + 3bd + d^2 &= 0 \end{aligned}$$

9. If $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of straight lines, prove that the square of the distance of their point of intersection from the origin is $\frac{c(a+b) - f^2 - g^2}{ab - h^2}$.

Solution: Given equation is $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ (1)

Let the lines represented by (1) be $l_1x + m_1y + n_1 = 0$ and $l_2x + m_2y + n_2 = 0$. So that

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = (l_1x + m_1y + n_1)(l_2x + m_2y + n_2) = 0$$

Equating the corresponding coefficients, we get

$$l_1l_2 = a, \quad m_1m_2 = b, \quad n_1n_2 = c, \quad m_1n_2 + m_2n_1 = 2f, \quad (2)$$

$$n_1l_2 + n_2l_1 = 2g, \quad l_1m_2 + l_2m_1 = 2h$$

Now solving $l_1x + m_1y + n_1 = 0$ and $l_2x + m_2y + n_2 = 0$, we get

$$\frac{x}{m_1n_2 - m_2n_1} = \frac{y}{n_1l_2 - n_2l_1} = \frac{1}{l_1m_2 - l_2m_1}$$

$$\begin{aligned} \text{Therefore, compute } x &= \frac{m_1n_2 - m_2n_1}{l_1m_2 - l_2m_1} = \frac{\sqrt{(m_1n_2 + m_2n_1)^2 - 4m_1m_2n_1n_2}}{\sqrt{(l_1m_2 + l_2m_1)^2 - 4l_1l_2m_1m_2}} = \frac{\sqrt{4f^2 - 4bc}}{\sqrt{4h^2 - 4ab}}, \text{ by (2)} \\ &= \sqrt{\frac{f^2 - bc}{h^2 - ab}} \end{aligned}$$

$$\text{Similarly, we can compute } y = \frac{n_1l_2 - n_2l_1}{l_1m_2 - l_2m_1} = \sqrt{\frac{g^2 - ac}{h^2 - ab}}$$

Thus, $\left(\sqrt{\frac{f^2 - bc}{h^2 - ab}}, \sqrt{\frac{g^2 - ac}{h^2 - ab}} \right)$ is the point of intersection of $l_1x + m_1y + n_1 = 0$ and

$$l_2x + m_2y + n_2 = 0.$$

Hence the square of the distance of the point of intersection from the origin is

$$\begin{aligned} &\left(\sqrt{\frac{f^2 - bc}{h^2 - ab}} \right)^2 + \left(\sqrt{\frac{g^2 - ac}{h^2 - ab}} \right)^2 = \frac{f^2 - bc + g^2 - ac}{h^2 - ab} \\ &= \frac{c(a+b) - f^2 - g^2}{ab - h^2} \text{ (Proved)} \end{aligned}$$

CHANGE OF AXES

Example 1

Transform the following equations to parallel axes through the new origin $(2, -4)$.

(i) $x^2 - 3y^2 + 4x + 6y = 0$

(iii) $3y^2 - 7x + 24y + 62 = 0$

(ii) $2x + 3y + 8 = 0$

(iv) $x^2 + y^2 - 4x + 8y - 17 = 0$.

Example 2

Transform the equation $x^2 - y^2 - 2\sqrt{2}x - 10\sqrt{2}y + 2 = 0$ when the axes are turned through $\frac{\pi}{4}$, the axes of coordinates being rectangular.

Example 3

Transform the equation $11x^2 + 24xy + 4y^2 - 20x - 40y - 5 = 0$ to rectangular axes through the point $(2, -1)$ and inclined at an angle $\tan^{-1}\left(-\frac{4}{3}\right)$.

Example 4

Transform the equation $17x^2 + 18xy - 7y^2 - 16x - 32y - 18 = 0$ to one in which there is no term involving x , y and xy .

Example 5

Transform the equation $x^2 - y^2 = a^2$ when the axes are turned through $\frac{\pi}{4}$.

Example 6

Through what angle must the axes be rotated to remove the term containing xy in $11x^2 + 4xy + 14y^2 = 5$? What is the transformed equation?

Example 7

By shifting the origin and rotating the axes of coordinates the equation $x^2 + 3xy + y^2 + 2x + 3y - 1 = 0$ is transformed into one of the form $ax^2 + by^2 = c$. Find the angle of rotation and the coordinates of the new origin.

Example 8

Find the angle through which the axes are to be turned so that the equation $\sqrt{3}x + y - 4 = 0$ may be reduced to the form $x = k$. Also, determine the value of k .

PAIR OF STRAIGHT LINES

Example 1

Show that the following equations represent a pair of straight lines. Find also their point of intersection and the angle between them.

- (i) $6y^2 - xy - x^2 + 30y + 36 = 0$.
- (ii) $x^2 - 5xy + 4y^2 + x + 2y - 2 = 0$.
- (iii) $3y^2 - 8xy - 3x^2 - 29x + 3y - 18 = 0$.

Example 2

Show that the equation $2x^2 + 7xy + 3y^2 + 8x + 14y + 8 = 0$ represents a pair of straight lines and find them. Find also their point of intersection.

Example 3

Prove that the equation $12x^2 + 7xy - 10y^2 + 13x + 45y - 35 = 0$ represents a pair of straight lines and find them. Find their point of intersection and the angle between them.

(Ref. S. L. Loney, page 97)

Example 4

Find the values of k so that the following equations represent a pair of straight lines. Also, find the angle between the lines.

- (i) $3x^2 + 10xy + 8y^2 - kx - 26y + 21 = 0$.
- (ii) $6x^2 + 11xy - 10y^2 + x + 31y + k = 0$.
- (iii) $12x^2 + kxy + 2y^2 + 11x - 5y + 2 = 0$.
- (iv) $6x^2 + xy + ky^2 - 11x + 43y - 35 = 0$.

Example 5

If the equation $ax^2 + 3xy - 2y^2 - 5x + 5y + c = 0$ represents a pair of straight lines perpendicular to each other, find the value of a and c .

Example 6

Write the equation of the pair of straight lines joining the origin to the points of intersection of the line $lx + my = 1$ with the curve $ax^2 + by^2 = 1$.

Example 7

Find the equation of the bisectors of the angles between the lines given by the equation $6x^2 + 7xy - 5y^2 - 22x - 41y - 8 = 0$.

Example 8

Prove that the equation $x^2 + 6xy + 9y^2 + 4x + 12y - 5 = 0$ represents two parallel lines and find the distance between them.

Example 9

Find the equations of the lines represented by the equation $abx^2 + (a^2 + b^2)xy + aby^2 + ab(a - b)(x - y) - a^2b^2 = 0$ and show that they are equidistant from the origin.

Example 10

Prove that the equation of the pair of straight lines passing through the origin and perpendicular to the lines given by $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ is $bx^2 - 2hxy + ay^2 = 0$.

Example 11

If $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of straight lines, prove that the area of the triangle formed by their bisectors and the x-axis is $\frac{\sqrt{(a-b)^2 + 4h^2}}{2h} \left(\frac{ca - g^2}{ab - h^2} \right)$.

Example 12

Under which condition one of the lines represented by the equation $ax^2 + 2hxy + by^2 = 0$ may coincide with one of the lines represented by $a'x^2 + 2h'xy + b'y^2 = 0$.

Example 13

Find the angle between the lines joining the origin to the points of intersection of the line $y - 3x - 2 = 0$ with the curve $x^2 + 2xy + 3y^2 + 4x + 8y - 11 = 0$.

Example 14

Prove that the pair of lines joining the origin to the points of intersection of the curve $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ by the $lx + my + n = 0$ are coincident if $a^2l^2 + b^2m^2 = n^2$.

Example 15

If $ax^2 + 2hxy + by^2 = 0$ be the combined equation of two adjacent sides of a parallelogram and the line $lx + my = 1$ be one of the diagonals, find the equation of the other diagonal.

THE CIRCLE

Example 1

Find the pole of the straight line $2x - y = 6$ with respect to the circle $5x^2 + 5y^2 = 9$.

Example 2

Show that the equation to the pair of tangents drawn from the origin to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is $c(x^2 + y^2) = (gx + fy)^2$.

Example 3

Prove that if the polar of a point P with respect to the circle $x^2 + y^2 = 37$ touches the circle $(x - 3)^2 + (y + 2)^2 = 25$ then the locus of P must be a conic.

Example 4

Find the equation of the direct common tangents to the circles $x^2 + y^2 = 16$ and $x^2 + y^2 + 6x - 8y = 0$.

Example 5

Find the equation of the transverse common tangents to the circles $x^2 + y^2 - x - 2y + 1 = 0$ and $x^2 + y^2 - 3x + 4y = 0$.

Example 6

Find the locus of the middle points of the chords of $x^2 + y^2 = a^2$ which subtend a right angle at the centre.

Example 7

From any point on one given circle, tangents are drawn to another given circle. Prove that the locus of the middle-point of chord of contact is a third circle.

Example 8

Find the equation of the circle passing through (5,7) and (6,6) and has its centre on $2x - y + 1 = 0$.

Example 9

Find the equations of the tangents to the circle $x^2 + y^2 - 6x + 4y = 12$ which are parallel to the straight line $4x + 3y + 5 = 0$.

Example 10

Show that the straight line $3x + 4y + 20 = 0$ touches the circle $x^2 + y^2 = 16$ and find the point of contact.

Example 11

The equation of a chord of the circle $x^2 + y^2 + 4x - 6y = 0$ is $x + 2y = 0$. Find the equation of the circle described on this chord as diameter.

Example 12

Find the length of the chord intercepted by the circle $x^2 + y^2 = a^2$ on the line $y = mx + c$.

Example 13

Find the equation of the circle which passes through the origin and makes intercepts a and b on the coordinate axes.

Example 14

$ABCD$ is a square with side of length ' l '. If AB and AD be taken as coordinate axes, show that the circle circumscribing the square will be $x^2 + y^2 = l(x + y)$.

Example 15

Show that the locus of the foot of the perpendicular from the origin to the line which always passes through a fixed point (h, k) is a circle.

Example 16

Find the direct and transverse common tangents to the circle $x^2 + y^2 - 22x + 4y + 100 = 0$ and $x^2 + y^2 + 22x - 4y - 100 = 0$.

Example 17

The polar of the point P with respect to the circle $x^2 + y^2 - 2bx = 0$ touches the circle $x^2 + y^2 = a^2$. Find the locus of P .

Example 18

Find the pole of the line $x + 2y + 1 = 0$ with respect to the circle $x^2 + y^2 - 4x + 6y - 11 = 0$.

Example 19

Find the equation of the circle whose diameter is the common chord of the circle $x^2 + y^2 + 2x + 3y + 1 = 0$ and $x^2 + y^2 + 4x + 3y + 2 = 0$.

SYSTEM OF CIRCLES

Example 1

Find the angle of intersection of the circles $x^2 + y^2 + 4x - 5 = 0$ and $x^2 + y^2 + 4x - 14y + 28 = 0$.

Example 2

Show that the circles $x^2 + y^2 - 4x - 6y + 4 = 0$ and $x^2 + y^2 - 10x - 14y + 58 = 0$ are orthogonal to each other.

Example 3

Find the equation of the circle which passes through (1, 1) and cuts orthogonally each of the circles $x^2 + y^2 - 8x - 2y + 16 = 0$ and $x^2 + y^2 - 4x - 4y - 1 = 0$.

Example 4

Find the equation of the circle which cuts orthogonally the circles $x^2 + y^2 + 4x + 7 = 0$, $2x^2 + 2y^2 + 3x + 5y + 9 = 0$ and $x^2 + y^2 + y = 0$.

Example 5

Find the equation of the circles which intersect $x^2 + y^2 - 6y + 1 = 0$ and $x^2 + y^2 - 4y + 1 = 0$ orthogonally and touch the line $3x + 4y + 5 = 0$.

Example 6

Find the equation of the circle which passes through (3, -2) and coaxial with each of the circles $x^2 + y^2 + 4x - 3y - 7 = 0$ and $x^2 + y^2 + 3x - 2y - 10 = 0$.

Example 7

Find the limiting points of the system of circles coaxial with the circles $x^2 + y^2 - 6x - 6y + 4 = 0$ and $x^2 + y^2 - 2x - 4y + 3 = 0$.

Example 8

Prove that the two circles each of which passes through (0, k) and (0, -k) and touch the line $y = mx + b$ will cut orthogonally if $b^2 = k^2(2 + m^2)$.

Example 9

Find the equation of the circle which passes through the origin and belongs to the coaxial system whose limiting points are (1, 2) and (3, 4).

Example 10

Find the radical axis of the coaxial system of circles whose limiting points are (-1, 2) and (2, 3).

Example 11

Prove that the limiting points of the system $x^2 + y^2 + 2gx + c + \lambda(x^2 + y^2 + 2fy + k) = 0$ subtend a right angle at the origin if $\frac{c}{g^2} + \frac{k}{f^2} = 2$.